

Time Series — Solutions

Practice Exam — Week 7: Spectral Density & Spectral Representation

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Problem 1 (5 points)

Statement.

For a discrete-time stationary process $\{X_t\}_{t \in \mathbb{Z}}$ with summable ACVS $\{\gamma_\tau\} \in \ell^1$ (sampling interval $\Delta = 1$), the spectral density function (SDF) is $\mathcal{S}(f) = \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i f \tau}$, with inverse $\gamma_\tau = \int_{-1/2}^{1/2} \mathcal{S}(f) e^{2\pi i f \tau} df$.

(a) Prove from the definition that any SDF is real-valued and even, $\mathcal{S}(-f) = \mathcal{S}(f)$, and write it in the cosine form $\mathcal{S}(f) = \gamma_0 + 2 \sum_{\tau \geq 1} \gamma_\tau \cos(2\pi f \tau)$. (1.5 pts)

(b) Show that $\gamma_0 = \text{Var}(X_t) = \int_{-1/2}^{1/2} \mathcal{S}(f) df$, and use this to interpret $\mathcal{S}(f)$ as a distribution of power across frequency. (1.5 pts)

(c) Take $\{X_t\} = \{\varepsilon_t\}$ (white noise of variance σ^2). Compute $\mathcal{S}(f)$ directly from its ACVS and explain in one sentence why the result earns the name “white”. (2 pts)

Solution.

(a) The ACVS of a real process is real and symmetric, $\gamma_{-\tau} = \gamma_\tau$. Pairing the term at lag τ with the one at lag $-\tau$ in the defining sum,

$$\begin{aligned} \mathcal{S}(f) &= \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i f \tau} = \gamma_0 + \sum_{\tau \geq 1} \gamma_\tau (e^{-2\pi i f \tau} + e^{+2\pi i f \tau}) \\ &= \gamma_0 + 2 \sum_{\tau \geq 1} \gamma_\tau \cos(2\pi f \tau), \end{aligned}$$

using $e^{-ix} + e^{ix} = 2 \cos x$. The right-hand side is a sum of real cosines, hence **real-valued**. It is **even** because $\cos$ is even: replacing $f \mapsto -f$ leaves every term unchanged, so $\mathcal{S}(-f) = \mathcal{S}(f)$. (Equivalently, $\mathcal{S}(-f) = \sum_{\tau} \gamma_\tau e^{2\pi i f \tau} = \sum_{\tau} \gamma_{-\tau} e^{-2\pi i f \tau} = \mathcal{S}(f)$ after re-indexing $\tau \mapsto -\tau$ and using symmetry.)

(b) Evaluate the inversion formula $\gamma_\tau = \int_{-1/2}^{1/2} \mathcal{S}(f) e^{2\pi i f \tau} df$ at lag $\tau = 0$, where the exponential equals 1:

$$\gamma_0 = \int_{-1/2}^{1/2} \mathcal{S}(f) e^0 df = \int_{-1/2}^{1/2} \mathcal{S}(f) df.$$

Since $\gamma_0 = \text{Cov}(X_t, X_t) = \text{Var}(X_t)$ (the process has mean 0), this is the spectral form of Parseval: the *total variance* of the process equals the integral of the SDF over the Nyquist band. Because $\mathcal{S} \geq 0$ (it is the transform of a non-negative-definite sequence), $\mathcal{S}(f) df$ acts like the “amount of variance/power carried by frequencies in $[f, f + df]$ ”: the SDF **distributes the total power** $\text{Var}(X_t)$ **across frequency**, with peaks marking the bands that dominate the process’s fluctuations.

(c) White noise is uncorrelated, so $\gamma_\tau = \sigma^2$ at $\tau = 0$ and $\gamma_\tau = 0$ for $\tau \neq 0$. Only the $\tau = 0$ term survives the sum:

$$\mathcal{S}(f) = \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i f \tau} = \gamma_0 e^0 = \boxed{\sigma^2}, \quad \text{for all } f.$$

The spectrum is **flat**: every frequency carries exactly the same power σ^2 , by analogy with white light containing all colours in equal measure — hence “white” noise. (Check: $\int_{-1/2}^{1/2} \sigma^2 df = \sigma^2 = \gamma_0 = \text{Var}(\varepsilon_t)$, consistent with part (b).)

Problem 2 (5 points)

Statement.

This problem tests whether a candidate sequence is a valid autocovariance.

(a) (The validity criterion.) State the criterion that decides whether a summable symmetric sequence $\{\gamma_\tau\}$ is the ACVS of some stationary process, expressed through its Fourier transform $\mathcal{S}$. Why is non-negativity of $\mathcal{S}$ the right test? (1 pt)

(b) (A truncated ACVS.) Let $\gamma_\tau = 1$ for $|\tau| \leq K$ and $\gamma_\tau = 0$ for $|\tau| > K$, where K is a positive integer. Compute $\mathcal{S}(f)$ in closed form (a Dirichlet kernel), then evaluate it at the Nyquist endpoint $f = \frac{1}{2}$ to decide for which K the sequence fails to be a valid ACVS. (2 pts)

(c) (MA(1)-shape ACVS.) Let $\gamma_0 = 1$, $\gamma_{\pm 1} = \rho$, and $\gamma_\tau = 0$ for $|\tau| \geq 2$, with $\rho \in \mathbb{R}$. Compute $\mathcal{S}(f)$ and determine the exact set of ρ for which $\{\gamma_\tau\}$ is a valid ACVS. Relate the answer to the largest possible lag-1 autocorrelation of an MA(1). (2 pts)

Solution.

(a) A summable symmetric sequence $\{\gamma_\tau\}$ is the ACVS of some stationary process **iff it is non-negative definite**, which (in the summable case) is equivalent to its Fourier transform being non-negative everywhere:

$$\mathcal{S}(f) = \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i f \tau} \geq 0 \quad \text{for all } f \in [-\frac{1}{2}, \frac{1}{2}].$$

This is the right test because, by Herglotz's theorem, every ACVS is non-negative definite and admits an integrated spectrum; in the summable case that spectrum has a density $\mathcal{S}$, and a genuine spectral density — being the limit of the (non-negative) expected periodogram, equivalently $\text{Var}(dZ(f))/df \geq 0$ in the spectral representation — can never be negative. So a single frequency at which $\mathcal{S}(f) < 0$ *disqualifies* the candidate.

(b) The sequence is summable and symmetric, so we may form

$$\mathcal{S}(f) = \sum_{\tau=-K}^K e^{-2\pi i \tau f} = 1 + 2 \sum_{\tau=1}^K \cos(2\pi \tau f) = \frac{\sin((2K+1)\pi f)}{\sin(\pi f)},$$

the Dirichlet kernel (the closed form follows from summing the geometric series $\sum_{\tau=-K}^K e^{-2\pi i \tau f}$). Evaluate at the Nyquist endpoint $f = \frac{1}{2}$, where $\cos(\pi \tau) = (-1)^\tau$. The alternating sum $\sum_{\tau=1}^K (-1)^\tau$ equals 0 when K is even and -1 when K is odd, so

$$\mathcal{S}(\frac{1}{2}) = 1 + 2 \sum_{\tau=1}^K (-1)^\tau = \begin{cases} 1, & K \text{ even,} \\ -1, & K \text{ odd.} \end{cases}$$

Thus for every **odd** K (write $K = 2q + 1$) we have $\mathcal{S}(\frac{1}{2}) = -1 < 0$, so the SDF goes negative and $\{\gamma_\tau\}$ is **not** a valid ACVS. (For even K this endpoint is harmless, but other frequencies of the Dirichlet kernel still dip below 0, so a flat truncated ACVS is in general not valid — truncating a genuine ACVS breaks non-negative-definiteness.)

(c) This is the ACVS shape of an MA(1). With only lags 0, ± 1 nonzero,

$$\mathcal{S}(f) = \gamma_0 + 2\gamma_1 \cos(2\pi f) = 1 + 2\rho \cos(2\pi f).$$

Since $\cos(2\pi f)$ ranges over $[-1, 1]$, the minimum of $\mathcal{S}$ is $1 - 2|\rho|$, attained at $f = \frac{1}{2}$ if $\rho > 0$ and at $f = 0$ if $\rho < 0$. Non-negativity $\mathcal{S}(f) \geq 0$ for all f therefore holds iff $1 - 2|\rho| \geq 0$, i.e.

$$\boxed{|\rho| \leq \frac{1}{2}}.$$

For $|\rho| > \frac{1}{2}$ the SDF dips below zero and the sequence is *not* a valid ACVS. This matches the known fact that an MA(1), $X_t = \varepsilon_t + \vartheta \varepsilon_{t-1}$, has lag-1 autocorrelation $\rho_1 = \vartheta/(1 + \vartheta^2)$, whose modulus is maximised at $|\vartheta| = 1$ giving $|\rho_1| = \frac{1}{2}$: **no** MA(1) (indeed no stationary process at all with this ACVS shape) can have $|\rho_1| > \frac{1}{2}$.

Problem 3 (5 points)

Statement.

Let $\{X_t\}$ be the MA(q) process $X_t = \sum_{j=0}^q \theta_j \varepsilon_{t-j}$ with $\theta_0 = 1$ and ε_t white noise of variance σ^2 .

(a) Recall (or derive) the ACVS $\gamma_\tau = \sigma^2 \sum_{j=0}^q \theta_j \theta_{j+|\tau|}$ for $|\tau| \leq q$ (and 0 otherwise), writing it in the double-sum form $\gamma_\tau = \sigma^2 \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k \mathbf{1}_{\{\tau=k-j\}}$. (1.5 pts)

(b) Starting from this double sum, derive the factored SDF

$$\mathcal{S}(f) = \sigma^2 \left| \sum_{j=0}^q \theta_j e^{-2\pi i j f} \right|^2,$$

and explain in one sentence why this form makes $\mathcal{S}(f) \geq 0$ automatic. (2 pts)

(c) Specialise to the MA(1) $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$. Give $\mathcal{S}(f) = \sigma^2(1 + \theta^2 - 2\theta \cos(2\pi f))$, and for $\theta > 0$ locate the frequency of the maximum and of the minimum of $\mathcal{S}$, giving the two extreme values. (1.5 pts)

Solution.

(a) Write $X_t = \sum_{j=0}^q \theta_j \varepsilon_{t-j}$ (with $\theta_0 = 1$). Since ε is mean-zero, so is X_t , and by bilinearity of covariance together with $\mathbb{E}[\varepsilon_{t-j} \varepsilon_{t+\tau-k}] = \sigma^2 \mathbf{1}_{\{t-j=t+\tau-k\}} = \sigma^2 \mathbf{1}_{\{k-j=\tau\}}$,

$$\gamma_\tau = \text{Cov}(X_t, X_{t+\tau}) = \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k \mathbb{E}[\varepsilon_{t-j} \varepsilon_{t+\tau-k}] = \sigma^2 \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k \mathbf{1}_{\{\tau=k-j\}}.$$

Collapsing the indicator (set $k = j + \tau$) gives the single-sum form

$$\gamma_\tau = \sigma^2 \sum_{j=0}^q \theta_j \theta_{j+|\tau|} \quad (|\tau| \leq q), \quad \gamma_\tau = 0 \quad (|\tau| > q),$$

where we used $\theta_j = 0$ for $j > q$; the value depends only on $|\tau|$ because the double sum is symmetric in $j \leftrightarrow k$. So the ACVS is finitely supported on $|\tau| \leq q$, hence trivially in ℓ^1 and the SDF exists.

(b) Transform the ACVS term by term, keeping the double-sum form so the indicator selects $\tau = k - j$:

$$\begin{aligned} \mathcal{S}(f) &= \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i \tau f} = \sigma^2 \sum_{\tau \in \mathbb{Z}} \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k \mathbf{1}_{\{\tau=k-j\}} e^{-2\pi i \tau f} \\ &= \sigma^2 \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k e^{-2\pi i (k-j)f} \\ &= \sigma^2 \left(\sum_{j=0}^q \theta_j e^{+2\pi i j f} \right) \left(\sum_{k=0}^q \theta_k e^{-2\pi i k f} \right) \\ &= \sigma^2 \left(\sum_{k=0}^q \theta_k e^{-2\pi i k f} \right) \left(\sum_{k=0}^q \theta_k e^{-2\pi i k f} \right) = \boxed{\sigma^2 \left| \sum_{j=0}^q \theta_j e^{-2\pi i j f} \right|^2}. \end{aligned}$$

In the third line the sum over τ collapsed the indicator, and $e^{-2\pi i (k-j)f} = e^{+2\pi i j f} e^{-2\pi i k f}$ factored the double sum into a number times its conjugate (the θ_j are real). The SDF is the squared

modulus of the MA transfer function $\Theta(e^{-2\pi if}) = \sum_j \theta_j e^{-2\pi i j f}$; **a squared modulus is ≥ 0 automatically**, so this form makes the necessary non-negativity of an SDF manifest.

(c) For the MA(1), $\Theta(z) = 1 - \theta z$, so $\sum_j \theta_j e^{-2\pi i j f} = 1 - \theta e^{-2\pi i f}$ and

$$\mathcal{S}(f) = \sigma^2 |1 - \theta e^{-2\pi i f}|^2 = \sigma^2 (1 - \theta e^{-2\pi i f})(1 - \theta e^{+2\pi i f}) = \sigma^2 (1 + \theta^2 - 2\theta \cos(2\pi f)),$$

matching the stated form (and the direct computation from $\gamma_0 = \sigma^2(1 + \theta^2)$, $\gamma_{\pm 1} = -\sigma^2\theta$). For $\theta > 0$ the term $-2\theta \cos(2\pi f)$ is smallest (most negative) when $\cos(2\pi f) = +1$, i.e. at $f = 0$, and largest when $\cos(2\pi f) = -1$, i.e. at $f = \frac{1}{2}$. Hence

$$\min_f \mathcal{S} = \mathcal{S}(0) = \sigma^2(1 + \theta^2 - 2\theta) = \sigma^2(1 - \theta)^2 \quad (\text{at } f = 0),$$

$$\max_f \mathcal{S} = \mathcal{S}(\frac{1}{2}) = \sigma^2(1 + \theta^2 + 2\theta) = \sigma^2(1 + \theta)^2 \quad (\text{at } f = \frac{1}{2}).$$

So a positive θ gives a **high-pass** shape: power is suppressed at low frequency ($f = 0$) and amplified at the Nyquist frequency ($f = \frac{1}{2}$). (For $\theta < 0$ the two extremes swap, giving a low-pass spectrum; and $\mathcal{S} \geq 0$ throughout, touching 0 at one frequency only when $|\theta| = 1$.)

Problem 4 (5 points)

Statement.

(Revision of Week 6: Fourier theory — convolution & aliasing.) Work with the continuous-time Fourier transform $G(f) = \int_{-\infty}^{\infty} g(t) e^{-2\pi i f t} dt$ and convolution $(g * h)(t) = \int_{-\infty}^{\infty} g(t - y) h(y) dy$ ($g, h \in L^1$).

(a) (Convolution theorem.) Prove that if $u = g * h$ then $U(f) = G(f) H(f)$, justifying carefully the interchange of integration (state the Fubini hypothesis you use). (2 pts)

(b) (Aliasing.) Let $g \in L^1$ be continuous with transform $\mathcal{G}$, and let G denote the transform of the sampled sequence $\{g(t)\}_{t \in \Delta\mathbb{Z}}$. Show the aliasing identity $G(f) = \sum_{k \in \mathbb{Z}} \mathcal{G}(f + k/\Delta)$, and define the Nyquist frequency. (2 pts)

(c) (A concrete alias.) A pure cosine of frequency $f_0 = 0.8$ Hz is sampled at rate $1/\Delta = 1$ Hz (so $\Delta = 1$ s). To which frequency in the principal Nyquist band $[-\frac{1}{2}, \frac{1}{2}]$ does f_0 fold, and what is the lowest sampling rate that would resolve f_0 without aliasing? (1 pt)

Solution.

(a) By definition $U(f) = \int_{-\infty}^{\infty} u(x) e^{-2\pi i x f} dx$ with $u(x) = \int g(x - y) h(y) dy$. Substitute and exchange the order of integration:

$$\begin{aligned} U(f) &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(x - y) h(y) dy \right) e^{-2\pi i x f} dx \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(x - y) e^{-2\pi i x f} dx \right) h(y) dy \quad (\text{Fubini}) \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(s) e^{-2\pi i s f} ds \right) e^{-2\pi i y f} h(y) dy \quad (s = x - y) \\ &= G(f) \int_{-\infty}^{\infty} h(y) e^{-2\pi i y f} dy = G(f) H(f). \end{aligned}$$

In the third line the substitution $s = x - y$ (for fixed y) splits the kernel as $e^{-2\pi i x f} = e^{-2\pi i s f} e^{-2\pi i y f}$, making the inner integral exactly $G(f)$, independent of y . The interchange of the two integrals is legitimate by **Fubini's theorem**, justified because the integrand is absolutely integrable:

$$\iint_{\mathbb{R}^2} |g(x - y) h(y)| dx dy = \int_{\mathbb{R}} |h(y)| \left(\int_{\mathbb{R}} |g(s)| ds \right) dy = \|g\|_1 \|h\|_1 < \infty,$$

since $g, h \in L^1$. Hence $u = g * h \iff U = G \cdot H$: convolution in time is multiplication in frequency.

(b) For $t \in \Delta\mathbb{Z}$ both inversion formulas give the *same* value $g(t)$ — the continuous one from $\mathcal{G}$, and the discrete-time (Fourier-series) one from G on one period:

$$g(t) = \int_{-\infty}^{\infty} \mathcal{G}(f) e^{2\pi i f t} df, \quad g(t) = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i f t} df.$$

Split the real line in the first integral into period-length blocks centred at k/Δ and shift each block by $-k/\Delta$:

$$\begin{aligned} g(t) &= \sum_{k \in \mathbb{Z}} \int_{k/\Delta - 1/(2\Delta)}^{k/\Delta + 1/(2\Delta)} \mathcal{G}(f) e^{2\pi i f t} df = \sum_{k \in \mathbb{Z}} \int_{-1/(2\Delta)}^{1/(2\Delta)} \mathcal{G}(f + k/\Delta) e^{2\pi i (f + k/\Delta) t} df \\ &= \sum_{k \in \mathbb{Z}} \int_{-1/(2\Delta)}^{1/(2\Delta)} \mathcal{G}(f + k/\Delta) e^{2\pi i f t} df, \end{aligned}$$

because $t = z\Delta$ for some $z \in \mathbb{Z}$ makes the extra phase $e^{2\pi i (k/\Delta) t} = e^{2\pi i k z} = 1$. Under the summability hypothesis $\sum_k \int |\mathcal{G}(f + k/\Delta)| df < \infty$ we may swap sum and integral:

$$\int_{-1/(2\Delta)}^{1/(2\Delta)} \left(\sum_{k \in \mathbb{Z}} \mathcal{G}(f + k/\Delta) \right) e^{2\pi i f t} df = g(t) = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i f t} df.$$

Equality of these Fourier-series coefficients for all $t \in \Delta\mathbb{Z}$ forces $G(f) = \sum_{k \in \mathbb{Z}} \mathcal{G}(f + k/\Delta)$ for a.e. f (and for all f , both sides being continuous). Sampling thus **folds** all the continuous content at the frequencies $\{f + k/\Delta\}$ onto the single frequency f . The **Nyquist frequency** is $f_N = 1/(2\Delta)$ (one half the sampling rate): content beyond $\pm f_N$ cannot be recovered from the samples.

(c) With $\Delta = 1$ s the Nyquist band is $[-\frac{1}{2}, \frac{1}{2}]$ Hz and the aliases of $f_0 = 0.8$ Hz are $f_0 + k = 0.8 + k$, $k \in \mathbb{Z}$. The one landing inside $[-\frac{1}{2}, \frac{1}{2}]$ is

$$0.8 + (-1) = -0.2 \text{ Hz},$$

so $f_0 = 0.8$ Hz **folds to** -0.2 Hz, i.e. appears at $|-0.2| = 0.2$ Hz (a cosine has a symmetric line pair, so it is observed as a 0.2 Hz oscillation — the wagon-wheel effect). To resolve f_0 without aliasing one needs the Nyquist frequency above f_0 : $1/(2\Delta) > 0.8$, i.e. a **sampling rate** $1/\Delta > 1.6$ **Hz** (any rate strictly above 1.6 samples per second).